
HybridKG: Reproducible Knowledge Graph Construction with Deterministic Backbone and LLM Enrichment

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Abstract

We introduce HybridKG, a hybrid framework for knowledge graph (KG) extraction from scientific text that combines the reproducibility of deterministic NLP with the semantic richness of large language models. Our approach separates KG construction into two phases: (1) a completely deterministic backbone using spaCy dependency parsing that extracts subject-verb-object triples and named entities with perfect reproducibility (Graph Edit Distance = 0.0 across runs), and (2) a constrained LLM enrichment phase using GPT-4o-mini that adds semantic relation typing, entity disambiguation, and implicit relation inference. We evaluate HybridKG on five arXiv papers and show that while the deterministic backbone produces only 2 relation types (direct action, prepositional), LLM enrichment expands this to 10+ semantic types (causal, temporal, definitional, procedural, attributional, comparative, etc.) at a cost of \$0.16 per paper. Our batched enrichment pipeline (50 entities/triples per call) makes the approach computationally practical. Code and data are available at <https://github.com/jack89-ML/hybrid-kg-paper>.

1 Introduction

Knowledge graphs (KGs) have become a fundamental data structure for information retrieval, question answering, and retrieval-augmented generation [6, 7]. However, constructing high-quality KGs from unstructured text remains challenging. Existing approaches fall into two categories, each with fundamental limitations.

Deterministic methods—such as OpenIE [1] and spaCy dependency parsing—are fully reproducible and computationally efficient. However, they produce only shallow syntactic relations (subject-verb-object triples) and lack the semantic richness needed for downstream applications. A spaCy dependency parser, for example, extracts “KGen—present-generator” as a direct action, but cannot determine whether the relation is causal, definitional, or procedural.

LLM-based methods [4, 8] produce semantically rich relations by prompting models like GPT-4 or Claude to extract entities and relations directly. However, these approaches suffer from two critical weaknesses: (1) they are non-reproducible—running the same prompt twice can produce different graphs—and (2) they are expensive, with per-paper costs often exceeding \$10 in API calls.

We propose HybridKG, a hybrid framework that combines the best of both worlds. Our key insight is that KG construction can be cleanly separated into two phases:

1. **Phase A — Deterministic Backbone:** A spaCy-based pipeline extracts SVO triples and NER entities with perfect reproducibility (proven via hash verification and GED=0.0 across random seeds).
2. **Phase B — Constrained LLM Enrichment:** A batched GPT-4o-mini pipeline adds semantic relation subtypes, entity disambiguation and categorization, and implicit relation inference—without creating new entities or relations from scratch.

Our contributions are:

- **A hybrid framework** that separates deterministic extraction from LLM enrichment, enabling reproducibility guarantees while achieving semantic richness.
- **Perfect reproducibility** of the deterministic backbone across 5 papers and 3 seeds each (GED=0.0, verified via SHA-256 hashing).
- **10+ semantic relation types** (causal, temporal, definitional, procedural, etc.) achieved at a cost of \$0.16 per paper—over 60x cheaper than purely LLM-based approaches.
- **A batched enrichment pipeline** that scales to large texts by processing entities and triples in batches of 50, reducing API latency from minutes per call to seconds.

2 Related Work

2.1 Deterministic KG Construction

Traditional KG construction methods rely on pattern matching and dependency parsing. OpenIE [1] extracts relation triples using syntactic patterns. spaCy’s dependency parser extracts subject-verb-object structures from constituency trees. While these methods are fully deterministic and computationally cheap, they produce only syntactic relations without semantic understanding.

2.2 LLM-based KG Construction

Recent work has explored using large language models for KG construction. GraphRAG [3] uses LLMs for graph-based RAG. KGGen [5] uses LLMs with iterative clustering for entity resolution. More broadly, works like Li et al. [4] and Zhu et al. [8] demonstrate LLM-based triple extraction. However, these methods are non-deterministic—the same prompt can yield different results across runs.

2.3 Hybrid Approaches

A growing body of work combines deterministic and neural methods. Some approaches use LLMs to refine or post-process extraction results. Our work differs in two key ways: (1) we enforce strict separation between deterministic extraction and LLM enrichment, preserving reproducibility guarantees, and (2) we use LLM enrichment only for *annotation* of existing relations, not creation of new ones.

3 Method

HybridKG operates in two distinct phases. Phase A is fully deterministic and establishes the backbone graph. Phase B uses a constrained LLM to enrich the backbone with semantic information.

3.1 Phase A: Deterministic Backbone

Given a text corpus, Phase A extracts a base KG using spaCy’s `en_core_web_lg` pipeline:

Entity Extraction. Named entities are extracted via spaCy’s NER module with categories including PERSON, ORG, GPE, DATE, CARDINAL, etc. Each entity is recorded with its text span and NER label.

Triple Extraction. Subject-Verb-Object (SVO) triples are extracted via dependency parsing. For each sentence, the root verb is identified, and its subject and object dependencies form the relation triple.

The entire Phase A is deterministic by construction. Given the same input text, spaCy produces identical parse trees, yielding identical entities and triples. We verify this via SHA-256 hashing of the serialized output.

3.2 Phase B: Constrained LLM Enrichment

Phase B operates on the existing entities and triples from Phase A. It uses GPT-4o-mini via the OpenRouter API with the following constraints:

- **No new entities:** The LLM can disambiguate and categorize existing entities but cannot create new ones.
- **No new triples:** The LLM can type existing relations but cannot create new subject-verb-object pairs.
- **Implicit relations only:** The LLM can infer multi-hop relations between existing entities.

Entity Enrichment. Each entity is processed in batches of 50. For each batch, the LLM receives a JSON array of entities and returns enriched versions with three additional fields: disambiguated (canonical name), category (e.g., Software, Concept, Method, Metric, Organization), and confidence.

Triple Enrichment. SVO triples are processed in batches of 50. The LLM assigns each triple a `relation_subtype` from a controlled vocabulary: causal, temporal, compositional, attributional, comparative, procedural, definitional, or other. A confidence score $[0,1]$ is also assigned.

Implicit Relation Inference. The LLM receives up to 100 triples and infers multi-hop relations (e.g., $A \rightarrow B$ and $B \rightarrow C$ implies $A \rightarrow C$). Relation types include: `contributes_to`, `part_of`, `specializes`, `contradicts`, `exemplifies`, `follows`, and `leads_to`.

3.3 Reproducibility Verification

We use Graph Edit Distance (GED) as the primary reproducibility metric:

$$\text{GED}(G_1, G_2) = 0.5 \cdot \frac{|\mathcal{N}_1 \triangle \mathcal{N}_2|}{|\mathcal{N}_1 \cup \mathcal{N}_2|} + 0.5 \cdot \frac{|\mathcal{E}_1 \triangle \mathcal{E}_2|}{|\mathcal{E}_1 \cup \mathcal{E}_2|} \quad (1)$$

where $\mathcal{N}$ and $\mathcal{E}$ are the node and edge sets respectively, and $\triangle$ denotes symmetric difference.

4 Experiments

4.1 Setup

We evaluate HybridKG on five arXiv papers spanning KG construction, graph RAG, and LLM-related surveys:

- **KGGen** [5]: Text-to-KG generation with language models
- **AriGraph** [2]: Graph-based memory for LLM agents
- **LLM-KGC Survey**: Survey of LLM-based KG construction
- **LLM-KG Roadmap**: Roadmap for LLMs and KGs
- **GraphRAG Survey**: Survey of graph-based RAG methods

For Experiment 1 (reproducibility), we process each paper in full with 3 random seeds. For Experiment 2 (LLM enrichment), we truncate to 30,000 characters per paper due to API token limits, running 1 seed. The LLM model is GPT-4o-mini via OpenRouter with temperature 0.1.

4.2 RQ1: Reproducibility (Phase A)

Table 1 shows the reproducibility results. Across all five papers and three seeds each, the deterministic backbone achieves $\text{GED} = 0.0$. SHA-256 hashing of serialized outputs confirms byte-level identical results across runs. This validates our claim that deterministic NLP provides a perfectly reproducible foundation.

Table 1: Reproducibility results for Phase A (deterministic backbone). All GED values are 0.0, demonstrating perfect reproducibility.

Paper	Nodes	Edges	GED
KGGen	775	263	0.000000
AriGraph	822	235	0.000000
LLM-KGC Survey	746	191	0.000000
LLM-KG Roadmap	2,280	468	0.000000
GraphRAG Survey	5,025	1,174	0.000000
Total	9,648	2,331	0.000000

Table 2: Comparison of deterministic backbone (B1, full text) vs. hybrid pipeline (B4, 30K chars). LLM enrichment adds semantic relation types, entity categories, and implicit relations.

Paper	B1 Nodes	B1 Edges	B4 Nodes	B4 Edges	Implicit	LLM Categories
KGGen	775	263	477	204	17	Concept, Software, Metric...
AriGraph	822	235	431	218	12	Organization, Location...
LLM-KGC Survey	746	191	340	180	13	Concept, Organization...
LLM-KG Roadmap	2,280	468	444	148	0	Concept, Event, Language...
GraphRAG Survey	5,025	1,174	536	152	0	Organization, Software...

4.3 RQ2: Semantic Enrichment (Phase B)

Table 2 compares the deterministic backbone (B1) with the hybrid pipeline (B4) on enrichment metrics. Note: B1 uses full text while B4 uses 30K chars (API limit), so node/edge counts are not directly comparable.

The key qualitative improvements are:

- **Relation types:** B1 produces only 2 types (direct_action, prepositional) from spaCy dependency parsing. B4 produces 10+ types including causal, temporal, definitional, procedural, attributional, and comparative.
- **Entity categories:** B1 uses only spaCy NER labels (PERSON, ORG, GPE...). B4 adds 20 LLM-generated categories such as Software, Concept, Method, Metric, and Facility.
- **Implicit relations:** B4 infers multi-hop relations that do not exist in the B1 backbone, providing richer graph connectivity.
- **Coverage:** Approximately 100% of B4 edges receive a semantic subtype, compared to 0% in B1.

4.4 RQ3: Cost Analysis

The total API cost for all five papers was \$0.82, or approximately \$0.16 per paper. Entity enrichment accounts for 40% of cost, triple enrichment 50%, and implicit inference 10%.

This compares favorably to purely LLM-based approaches, which typically require thousands of tokens per extraction step and can cost \$5–\$20 per paper for comparable quality.

4.5 Limitations

Our evaluation has several limitations. First, the B4 results use truncated text (30K chars) due to API token limits, while B1 uses full text. Second, we run only 1 seed for B4; reproducibility of the LLM enrichment phase is not measured. Third, LLM-assigned categories and relation types have not been validated by human annotators. Finally, our evaluation is limited to 5 papers; scaling to larger corpora would require additional validation.

5 Conclusion and Future Work

We presented HybridKG, a hybrid framework for KG construction that combines deterministic reproducibility with LLM-based semantic enrichment. Our experiments demonstrate perfect reproducibility of the deterministic backbone (GED=0.0) alongside significant semantic enrichment at minimal cost (\$0.16 per paper).

Future work includes: (1) human evaluation of LLM-assigned categories and relation types, (2) comparison with alternative baselines (B2: LLM zero-shot, B3: LLM schema-guided), (3) multi-seed B4 runs using cheaper models, and (4) validation on larger, more diverse corpora.

Broader Impact

This work enables reproducible KG construction with richer semantics, benefiting applications in scientific literature analysis, biomedical knowledge discovery, and document understanding. Potential risks include over-reliance on LLM annotations without human validation, and the computational cost of API-based enrichment at scale.

References

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